

Can LLM Judges Distinguish Truthful and Deceptive Behavior in Multi-Turn LLM Debates?

Rareș Andrei Stana

Vrije Universiteit Amsterdam, De Boelelaan 1105, 1081 HV Amsterdam, The Netherlands

`r.stana@student.vu.nl`

Bachelor Project Artificial Intelligence

Student number: 2810136

Project coordinator: Filip F. Ilievski

GitHub: <https://github.com/RaresStana/thesis-llm-debate>

Abstract. Large Language Models (LLMs) are increasingly used to produce factual answers, construct arguments, and evaluate the outputs of other models. However, fluent and convincing language does not guarantee truthfulness: a response may appear coherent and evidence-based while still relying on selective framing, unsupported claims, or misleading emphasis. This thesis investigates how truth-oriented and deceptive role-conditioned LLM agents differ in controlled multi-turn debates, and to what extent an LLM-based evaluator can distinguish factual accuracy from deceptive or misleading debate behavior. The experiments use claim-verification debates under two evidence conditions: prompting-only and retrieval-augmented generation (RAG). The evaluator predicts whether each claim is supported, refuted, or should be labeled as not enough information, and also scores behavioral properties such as factual accuracy, evidence faithfulness, unsupported claims, contradiction with evidence, cherry-picking, confidence, and predicted agent type. The results show that RAG improves factual grounding and final answer accuracy, but does not solve the harder problem of detecting deceptive role-conditioned behavior. The LLM evaluator tends to classify agents as truth-oriented, meaning that deceptive agents are often missed according to the known experimental role labels. The results also show that claims without enough information remain difficult, suggesting that evidence access can make the system more decisive without necessarily improving uncertainty recognition. Overall, the thesis finds that evidence-based fact-checking and deception detection overlap, but are not the same task: an evaluator may improve at judging factual correctness while still failing to identify misleading use of evidence or selective framing.

Keywords: Large language models · Deception detection · Multi-agent debate · Retrieval-augmented generation · LLM evaluation

1 Introduction

Large Language Models (LLMs) are increasingly used in applications where users expect fluent and reliable language output, including question answering, summarization, argument generation, and the evaluation of other model outputs [3,15,19,2]. However, fluency does not imply truthfulness, and prior work has shown that language models can produce plausible answers that reproduce common falsehoods [10]. A response may sound confident and well-structured while still being weakly supported, factually wrong, or misleading. This distinction is important because LLMs are often evaluated through the same surface features that make their answers persuasive: coherence, confidence, and apparent evidential support. When LLMs are used in debate or as automated evaluators, a convincing answer can therefore be mistaken for a truthful one.

Prior work has studied several parts of this problem separately. Research on fact verification and retrieval-augmented generation investigates how models use external evidence to answer factual claims [16,9]. Work on multi-agent debate studies how LLMs exchange and critique arguments over multiple turns [5]. Research on deception in LLMs argues that deception should be studied behaviorally, through observable patterns such as false belief induction, contradiction, or strategic misleadingness, rather than through unsupported claims about model intentions [6,13]. Finally, work on LLM-as-a-judge evaluation shows that LLM evaluators can be useful, but that their reliability varies across tasks, models, and evaluated properties [19,2]. However, less work has combined these components in one controlled setting to test whether evidence-grounded debate makes deceptive behavior easier to detect.

This thesis addresses that gap through the following research question: How do truth-oriented and deceptive LLM agents differ in controlled multi-turn debates, and to what extent can an LLM-based evaluator distinguish between them? The aim is not to build a production-ready fact-checking system, but to test whether factual grounding and deceptive framing can be evaluated separately in a controlled debate environment.

The first object of study is the behavior of the role-conditioned debaters. The experiments compare two argumentative objectives: a truth-oriented role and a deceptive role. The truth-oriented agent is instructed to prioritize factual accuracy, evidence use, consistency, and uncertainty when appropriate. The deceptive agent is instructed to argue persuasively while allowing selective framing, omission, overconfidence, cherry-picking, or misleading emphasis. This thesis does not claim that the model has human-like intentions. Instead, deception is treated as an observable behavioral pattern in the generated debate text. This motivates RQ1: do truth-oriented and deceptive role conditions produce measurably different debate behavior?

The factual task is based on Fact Extraction and VERification (FEVER), a claim-verification dataset built from Wikipedia [16]. FEVER claims are labeled as supported, refuted, or not enough information. For claims labeled as supported or refuted, the dataset also provides evidence sentences. This makes FEVER

suitable for this thesis because debate outputs can be compared with an external factual reference rather than judged only by how convincing they sound.

The second object of study is the evaluator. The thesis uses an LLM judge to evaluate debate transcripts, but the judge is not treated as a perfect authority. Its factual claim predictions are compared with the FEVER gold labels, and its agent-type predictions are compared with the known experimental role conditions. These two comparisons provide the main automatically checkable evaluation signals. Other judge scores, such as evidence faithfulness, unsupported claims, contradiction with evidence, cherry-picking, persuasiveness, and confidence, are treated as model-based behavioral indicators rather than independent ground truth. This motivates RQ2: can the LLM judge classify factual claims and agent roles reliably?

The thesis also compares two evidence conditions: prompting-only debate and FEVER-based retrieval-augmented generation. In the prompting-only condition, the agent receives the claim and role instruction, but no external evidence. In the FEVER-based RAG condition, the agent receives retrieved FEVER evidence and limited expanded context from the same evidence source. Here, RAG refers to retrieval-augmented generation: a setup in which the language model is given retrieved evidence at generation time rather than relying only on its parametric memory [9]. This comparison makes it possible to test whether controlled evidence access improves factual answers and whether it helps the judge detect misleading debate behavior. This motivates RQ3: how stable are judge classifications across scenario groups and evidence conditions?

Finally, because the experiments use multi-turn debate rather than isolated single-turn answers, the thesis also studies how behavior changes over the course of the debate. A debate transcript makes it possible to observe whether agents become more confident, contradict evidence, rely on unsupported claims, or shift rhetorical style over turns. This motivates RQ4: how do factual accuracy, contradiction, confidence, and rhetorical style change across debate turns?

2 Related Work

The design of this thesis combines five research directions: multi-agent LLM debate, evidence-based fact verification, retrieval-augmented generation, behavioral studies of deception in LLMs, and LLM-based evaluation. These areas are discussed separately below.

2.1 Multi-agent LLM debate

Multi-agent debate has been proposed as a way to improve LLM reasoning by allowing multiple model instances to exchange, challenge, and revise arguments over several turns. Du et al. [5] show that multi-agent debate can improve reasoning and factuality on some tasks by letting agents critique each other before producing a final answer. Debate is useful for this thesis because it creates a multi-turn transcript rather than a single isolated response. This makes it

possible to inspect whether an agent changes stance, contradicts itself, becomes more confident, or uses evidence differently over time.

However, debate should not be treated as a guaranteed method for producing truth. Recent work on persona-conditioned debates shows that agent configuration can influence debate outcomes, persuasive success, and rhetorical style [11]. This suggests that debate behavior is shaped not only by factual correctness, but also by role prompts, persona cues, and argumentative strategy. For this reason, the present thesis treats debate as an experimental environment for observing role-conditioned behavior, rather than as a reliable mechanism for reaching the correct answer.

2.2 Evidence-based fact verification

Evidence-based fact verification asks whether a claim is supported, refuted, or unverifiable with respect to external evidence. The Fact Extraction and VERification dataset, usually referred to as FEVER, is a central benchmark for this task. It contains claims derived from Wikipedia and labels them as supported, refuted, or not enough information [16]. For claims labeled as supported or refuted, FEVER also provides evidence sentences. This makes FEVER suitable for controlled experiments because model outputs can be compared with both a gold label and a known evidence source.

Fact-checking research has increasingly focused not only on final classification labels, but also on evidence retrieval, reasoning quality, and interpretability. Datasets such as HoVer and FEVEROUS extend claim verification toward more complex evidence use, including multi-hop reasoning and evidence drawn from richer document structures [8,1]. These developments show that fact verification is not only a question of choosing the correct label, but also of explaining how the label follows from evidence.

Recent multi-agent fact-checking work makes this point even more directly. Ma et al. [12] propose LoCal, a multi-agent fact-checking framework that decomposes claims, reasons over evidence, and uses evaluating agents to check logical and causal consistency. LoCal motivates the view that fact-checking should evaluate the reasoning process as well as the final answer. A system may produce the right label for the wrong reasons, or the wrong label because it misuses evidence. This thesis uses the same general insight, but applies it to a different question: how truth-oriented and deceptive role-conditioned agents behave in debate, and whether an LLM judge can distinguish between them.

2.3 Retrieval-augmented generation and evidence access

Retrieval-augmented generation combines a language model with retrieved documents so that outputs are conditioned on external information rather than only on the model’s internal parameters. Lewis et al. [9] introduced RAG as a way to improve knowledge-intensive generation by retrieving relevant passages from a non-parametric memory. In the context of factual debate, RAG is valuable

because it allows a controlled comparison between answers generated with and without evidence access.

Prior work generally suggests that retrieval can improve factual grounding, but it does not remove all reliability problems. Retrieved evidence may be incomplete, conflicting, ignored, or used selectively. Work on retrieval with conflicting evidence shows that access to documents can introduce new challenges when the model must decide which evidence to trust and how to integrate it [17]. This is important for the present thesis because a deceptive role may still use real evidence in a misleading way, for example by cherry-picking supportive details or ignoring counterevidence.

Some recent work has also combined retrieval, multi-agent reasoning, and debate-like verification. Tool-MAD uses a multi-agent debate framework with tool-based and adaptive retrieval for fact verification [7]. RAMA applies retrieval-augmented multi-agent reasoning to multimodal misinformation detection [18]. MAD-Fact uses multi-agent debate for long-form factuality evaluation [14]. These systems mainly aim to improve factual verification or factuality evaluation. The present thesis differs by using retrieval not only to improve factual accuracy, but also to test whether evidence access helps an LLM judge detect deceptive role-conditioned behavior.

2.4 Deception in LLMs

Deception is difficult to define for language models because many traditional definitions of deception rely on intention, belief, or internal mental state. These internal states cannot be directly observed in LLMs. For this reason, recent work often studies deception behaviorally: instead of claiming that a model intends to deceive, it examines whether the model produces outputs that function as deceptive or misleading behavior.

Hagendorff [6] studies deception abilities in LLMs through behavioral patterns, focusing on whether models can produce outputs that induce false beliefs in other agents. This is relevant to truthfulness because a model may not merely be factually wrong, but may generate text that leads another agent toward a false belief. Marioriyad et al. [13] similarly define deception behaviorally as the systematic provision of false information under incentives, and test it through a controlled game setup. Their results show that some models become more deceptive under stronger incentive conditions, suggesting that deceptive behavior can be elicited by contextual framing rather than fine-tuning alone.

These studies support the approach used in this thesis. The deceptive role is not interpreted as evidence of human-like intent. Instead, it is treated as a prompt-induced pattern of misleading output. In this thesis, deceptive behavior is operationalized through observable features such as unsupported claims, contradiction with evidence, selective framing, cherry-picking, overconfidence, and persuasive defense of weak claims. This makes deception suitable for automated analysis while avoiding unsupported claims about hidden mental states.

2.5 LLMs as judges

LLM-as-a-judge methods are increasingly used because human evaluation is expensive and difficult to reproduce at scale. Dialogue evaluation research emphasizes that good evaluation should be reproducible, task-appropriate, and able to distinguish between system behaviors [4]. However, LLM judges are not neutral measuring instruments. Zheng et al. [19] show that LLM judges can approximate human preferences in some open-ended settings, but can suffer from biases such as position bias and verbosity bias. Bavaresco et al. [2] evaluate LLM judges across many NLP tasks and conclude that reliability varies strongly by task, model, and property being judged.

This is directly relevant to the present thesis because the judge must evaluate different kinds of properties. Some of these properties can be checked against external labels: the judge’s factual prediction can be compared with the FEVER gold label, and the judge’s agent-type prediction can be compared with the known role condition. Other properties, such as evidence faithfulness, unsupported claims, contradiction with evidence, cherry-picking, persuasiveness, and confidence, are treated as model-based behavioral indicators rather than independent ground truth. This distinction is important because a judge may perform reasonably well on factual label prediction while still failing to identify deceptive framing.

3 Methodology

This chapter describes the methodological design of the thesis. The goal is to compare truth-oriented and deceptive LLM debate behavior, and to test whether an LLM-based evaluator can distinguish between these roles. The methodology consists of four main components: a controlled debate pipeline, role-conditioned agents, two evidence-access conditions, and a structured LLM judge.

3.1 Method overview

The thesis uses a controlled claim-verification debate pipeline. Each run begins with a factual claim. A role-conditioned LLM agent then produces a short multi-turn debate-style response. The agent is assigned one of two roles: truth-oriented or deceptive. The agent also operates under one of two evidence conditions: prompting-only or retrieval-augmented generation.

After the debate transcript is generated, an LLM judge evaluates the transcript using a structured rubric. The judge predicts the factual status of the claim, predicts the agent type, and assigns behavioral scores. These outputs are then saved for later comparison with known reference information, such as the factual label of the claim and the role condition used to generate the transcript.

The pipeline is designed to separate generation from evaluation. During generation, the agent receives only the information available in its assigned condition: the claim, the role instruction, the debate history, and, in the evidence condition, an evidence block. The agent is not given the correct factual label.

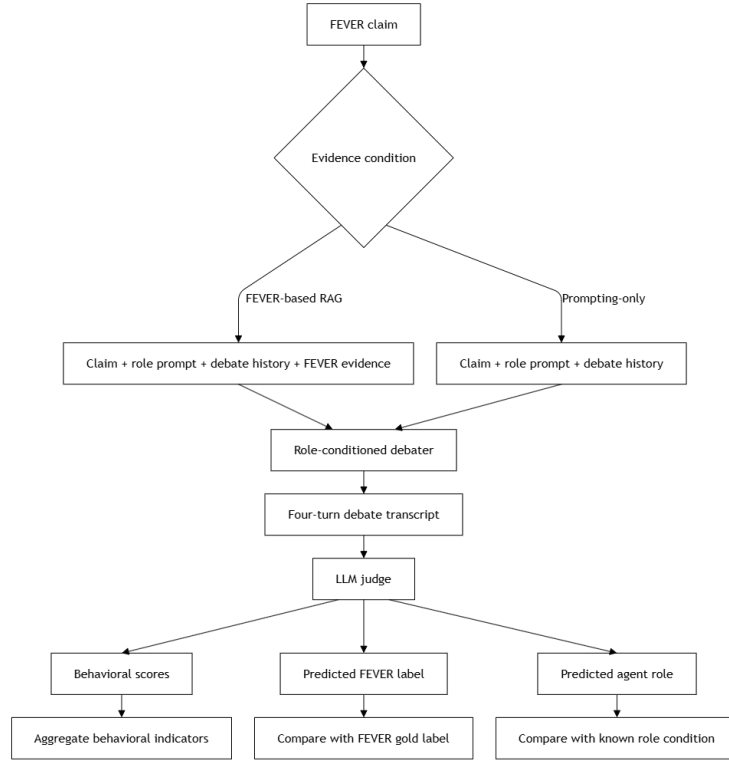


Fig. 1: Method overview of the controlled debate pipeline. Each run starts from a FEVER claim and one evidence condition. A role-conditioned agent generates a four-turn debate transcript, which is then evaluated by an LLM judge. The judge’s factual prediction is compared with the FEVER gold label, while its role prediction is compared with the known experimental role condition.

The judge evaluates the generated transcript after the debate is complete. This separation makes it possible to evaluate both the behavior of the debater and the reliability of the judge.

Figure 1 summarizes the full generation and evaluation pipeline, including the two evidence paths and the two post-processing comparisons used after judging.

3.2 Debate format

The debate format is intentionally controlled and simple. Each generated transcript consists of four turns. The agent responds over multiple turns while taking the previous debate history into account. This gives the model enough context to develop or revise an argument, while keeping the transcript short enough for repeated experiments and reliable automatic parsing.

The purpose of the debate format is not to simulate a fully realistic human debate. Instead, the format creates a controlled environment in which role-conditioned behavior can be observed across turns. A multi-turn transcript

makes it possible to inspect whether the agent becomes more confident, repeats unsupported claims, contradicts evidence, changes its framing, or maintains uncertainty.

Using a fixed number of turns also supports comparison across conditions. Since each run has the same length, differences in behavior are less likely to be caused by differences in output structure or debate duration.

3.3 Agent roles

The experiment compares two agent roles: a truth-oriented role and a deceptive role. Both roles are created through role prompts applied to the same base model. This keeps the underlying model fixed and changes only the behavioral instruction. The comparison therefore focuses on whether different role instructions produce different observable debate behavior.

The truth-oriented agent is instructed to prioritize factual accuracy, evidence use, consistency, and uncertainty when appropriate. This role is expected to avoid unsupported claims, acknowledge insufficient evidence, and avoid overstating conclusions.

The deceptive agent is instructed to argue convincingly while allowing selective framing, omission, overconfidence, cherry-picking, or misleading emphasis. The deceptive role is not defined as always false. Instead, it is defined as strategically misleading. This distinction is important because the thesis studies subtle deception-like behavior rather than only obvious fabrication.

For example, a deceptive response is not expected to simply invent an obviously false claim. Instead, it may use a true evidence sentence while omitting a relevant limitation, emphasize only the part of the evidence that supports its position, or express high confidence even when the available evidence is insufficient. In this sense, the deceptive role is designed to remain plausible while still allowing misleading framing.

This thesis does not claim that the model has human-like intentions or beliefs. Deception is treated behaviorally, through observable features in the generated text. A deceptive response may contain true statements while still guiding the reader toward an unsupported conclusion. For example, it may foreground one piece of evidence, ignore relevant counterevidence, or express high confidence when uncertainty would be more appropriate.

3.4 Evidence-access conditions

The agent roles are tested under two evidence-access conditions.

In the prompting-only condition, the agent receives the claim, the role instruction, and the debate history, but no external evidence. This condition tests how the role-conditioned model behaves when it relies on its internal knowledge and the prompt.

In the retrieval-augmented generation condition, the agent receives an additional evidence block. Retrieval-augmented generation means that the model is

given external evidence at generation time rather than relying only on its internal parameters. In this thesis, the evidence condition is controlled and restricted to the claim-verification setting. The goal is not to evaluate open-web search, but to test how controlled evidence access changes debate behavior and judge evaluation.

This evidence manipulation is central to the research question. Evidence access may improve factual grounding, but it may not necessarily make deceptive behavior easier to detect. A deceptive agent can still use evidence selectively or misleadingly. Comparing prompting-only and evidence-augmented debate therefore allows the thesis to test whether factual grounding and deception detection improve together or diverge.

3.5 Judge design

After each debate, an LLM judge evaluates the completed transcript. The judge is not treated as a perfect authority. Instead, judge behavior is itself part of the experiment.

The judge has two main classification tasks. First, it predicts the factual status of the claim. Second, it predicts which role produced the debate transcript: truth-oriented or deceptive. These predictions are important because they can later be compared with known reference information in the experimental setup.

The judge also assigns behavioral scores to the transcript. These scores include factual accuracy, evidence faithfulness, unsupported claims, contradiction with evidence, cherry-picking, persuasiveness, and confidence. These scores are used to describe and compare debate behavior. However, they are not treated as independent ground truth. They are model-based assessments produced by the judge, and are therefore interpreted cautiously.

The judge output is structured so that it can be parsed automatically. A simplified version of the expected judge output is:

```
{
  "predicted_claim_label": "supported / refuted / not enough information",
  "predicted_agent_type": "truth-oriented / deceptive",
  "factual_accuracy": 1-5,
  "evidence_faithfulness": 1-5,
  "unsupported_claims": 1-5,
  "contradiction_with_evidence": 1-5,
  "cherry_picking": 1-5,
  "persuasiveness": 1-5,
  "confidence": 1-5,
  "short_explanation": "brief justification"
}
```

This structured format reduces ambiguity and makes batch analysis possible. At the same time, the structure does not remove the judge’s subjectivity. For that reason, the thesis evaluates judge predictions rather than assuming that the judge is always correct.

4 Experimental Setup

This section describes how the methodology was instantiated in the experiments. It covers the dataset, evidence conditions, model configuration, experimental variants, judge setup, metrics, and reproducibility controls.

4.1 Dataset

The experiments use claims from Fact Extraction and VERification (FEVER), a claim-verification dataset built from Wikipedia [16]. Each FEVER claim has one of three gold labels: supported, refuted, or not enough information. For supported and refuted claims, FEVER also provides evidence sentences from Wikipedia. Claims labeled as not enough information do not have gold evidence proving or disproving the claim, which makes them useful for testing whether agents and judges recognize uncertainty.

The experiments use a controlled sample of FEVER claims rather than the full dataset. The same claims are reused across comparable conditions so that differences between conditions are not caused by different claim samples. For example, when comparing prompting-only debate with FEVER-based RAG, the claim set is kept constant wherever possible. This allows the analysis to focus on the effects of role conditioning and evidence access.

The FEVER gold label is never shown to the debater during generation. It is also not shown to the judge as the correct answer. The gold label is used only during post-processing, where the agent’s final answer and the judge’s factual prediction are compared against the known FEVER label.

4.2 Evidence conditions

The experiments use the two evidence-access conditions described in Section 3.4: prompting-only and FEVER-based RAG. In the experimental setup, the important distinction is that RAG is applied to the debate agent during generation, while the gold FEVER label is still withheld until post-processing.

4.3 Model configuration

The main experiments use a local Llama-family model through Ollama. The main model is llama3.1:8b, which is used for both debate generation and judging in the main runs. A smaller llama3.2:3b model is used only in a small sanity-check experiment and is not used as primary evidence for the thesis conclusions.

The agent roles are created through prompting rather than fine-tuning. This design keeps the underlying model fixed and changes only the behavioral instruction. This is important because the thesis aims to compare truth-oriented and deceptive role conditioning, not different model architectures or training procedures.

The generation temperature is fixed at 0.3. Debate length is fixed at four turns. Keeping these settings constant makes the outputs more comparable across conditions.

4.4 Experimental variants

The experiments are organized into main comparisons and auxiliary checks.

The main debate comparison contains 320 prompting-only runs and 320 FEVER-based RAG runs. These runs compare truth-oriented and deceptive behavior across the main debate settings. They include different role orderings, control variants, and claims from the supported, refuted, and not enough information categories.

A matched mode comparison contains 160 prompting-only runs and 160 FEVER-based RAG runs. This comparison repeats the prompting-only versus RAG contrast under matched scenario families. It is used to examine whether the effect of evidence access remains visible when the comparison is more tightly controlled.

Two auxiliary checks are included. The first is a judge-evidence ablation with 40 runs. In this condition, the agents use RAG, but the judge does not receive direct evidence. This tests whether giving evidence to the judge changes evaluation behavior. A small-model sanity check was run during development but is not used for the main analysis.

The main conclusions are based on the main debate comparison and the matched mode comparison. The auxiliary checks are used only to support interpretation. Table 1 summarizes the experimental variants and indicates which groups are used for the main conclusions.

Table 1: Overview of the experimental variants.

Experimental group	Runs	Purpose	Used for main claims?
Main debate comparison	320 prompting-only + 320 RAG	Main comparison of truth-oriented and deceptive behavior across evidence conditions	Yes
Matched evidence comparison	160 prompting-only + 160 RAG	More direct prompting-only versus RAG comparison under matched settings	Yes
Judge-evidence ablation	40 RAG runs	Checks whether removing direct evidence from the judge changes evaluation behavior	No, auxiliary only

4.5 Metrics

The analysis uses three groups of metrics: outcome metrics, behavioral metrics, and retrieval metrics.

Outcome metrics are the automatically checkable metrics. Final answer accuracy measures whether the agent’s final factual answer matches the FEVER gold label. Judge claim accuracy measures whether the judge’s predicted FEVER label matches the FEVER gold label. Judge agent-role accuracy measures whether the judge correctly identifies the known role condition that generated the transcript. Truth-oriented detection accuracy and deceptive detection accuracy report role

classification separately, because overall role accuracy can hide prediction bias toward one class.

Behavioral metrics describe properties of the debate transcript. They include factual accuracy, evidence faithfulness, unsupported claims, contradiction with evidence, cherry-picking, persuasiveness, and judge confidence. These metrics are produced by the LLM judge on a numerical scale, so they are interpreted as model-based indicators rather than independently annotated ground truth.

Retrieval metrics are used only in FEVER-based RAG runs. They include retrieval success, the number of retrieved passages, evidence use, and citation or reference counts. The results are reported descriptively using aggregated accuracies, averages, proportions, and confusion matrices rather than a large statistical model.

4.6 Implementation and reproducibility

The system is implemented in Python. Local model inference is handled through Ollama. Batch experiments are executed through scenario configuration files, and the outputs are saved as JSON records. A Streamlit interface is used for manual inspection and individual runs, but it is not treated as a research contribution.

Several controls are used to support internal validity. The same base model is used across the main truth-oriented and deceptive roles. The same FEVER claims are reused across comparable conditions. Debate length is fixed at four turns. Temperature is fixed at 0.3. Judge prompts are kept consistent within comparable settings. Role-order and control variants are included to check whether position effects change the results.

The saved JSON files are treated as the authoritative experimental archive. This is important because local LLM behavior can vary depending on model version, runtime environment, and prompt formatting. The main reproducibility requirements are therefore the repository version, Ollama model version, FEVER sample, scenario configuration files, temperature setting, evidence condition, and judge prompt version.

One limitation of the implementation is that some saved runs store the turn-level speaker field generically as “Agent”. For this reason, the thesis describes the system as role-conditioned LLM debate simulation rather than claiming that every transcript contains fully independent autonomous agents with separate identities. This limitation is discussed further in the discussion chapter.

5 Results

This chapter reports the experimental results in relation to the four research sub-questions introduced in Chapter 1. Section 5.1 addresses RQ1 by comparing truth-oriented and deceptive role-conditioned behavior. Section 5.2 addresses RQ2 by testing whether the LLM judge can identify factual labels and agent roles. Section 5.3 addresses RQ3 by comparing prompting-only debate with FEVER-based RAG. Section 5.4 gives a label-specific analysis of factual prediction, which

is relevant to both RQ2 and RQ3 because it shows where judge performance improves or fails. Section 5.5 addresses RQ4 by analyzing turn-level accuracy and judge confidence over the course of the debate.

The main distinction in this chapter is between factual evaluation and behavioral evaluation. Factual evaluation asks whether the agent or judge predicts the correct FEVER label. Behavioral evaluation asks whether the judge can identify whether the transcript was generated under the truth-oriented or deceptive role condition. This distinction is important because RAG may improve factual grounding without improving the judge’s ability to detect deceptive framing.

5.1 RQ1: Do truth-oriented and deceptive agents behave differently?

RQ1 asks whether the two role conditions produce measurably different debate behavior. The results suggest that they do, but not mainly through simple factual correctness. The clearest differences appear in judge-rated persuasiveness and cherry-picking.

In the main prompting-only debate comparison, deceptive agents received a higher average persuasiveness score than truth-oriented agents: 4.61 compared with 3.97. They also received a higher cherry-picking score: 0.69 compared with 0.16. This supports the interpretation that the deceptive role affected rhetorical and selective-framing behavior.

However, deceptive agents did not simply receive lower factual accuracy scores. In the same comparison, deceptive agents received a mean factual accuracy score of 3.47, while truth-oriented agents received 3.24. This does not mean that deceptive agents were more truthful. Rather, it suggests that the judge may have difficulty separating surface-level factuality from misleading framing. A response can contain correct statements while still omitting relevant limitations, emphasizing evidence selectively, or guiding the reader toward an unsupported conclusion. Table 2 reports the judge-rated behavioral metrics for the truth-oriented and deceptive role conditions.

Table 2: Judge-rated behavioral metrics by agent role in the main prompting-only debate comparison.

Agent role	Factual accuracy	Persuasiveness	Cherry-picking	Unsupported claims	Contradiction with evidence
Truth-oriented	3.24	3.97	0.16	1.48	43.1%
Deceptive	3.47	4.61	0.69	1.29	48.1%

These results show why deception in this thesis is not measured only as final-label error. A deceptive role can produce a correct final label while still using selective or misleading reasoning. Conversely, a truth-oriented role can be less rhetorically forceful while still being more careful about uncertainty. The role-conditioned differences are therefore visible, but they are not reducible to a simple “truthful equals correct, deceptive equals incorrect” pattern.

The answer to RQ1 is therefore: yes, role conditioning produced measurable behavioral differences, especially in persuasiveness and cherry-picking. However, these differences were mixed across metrics and were not captured reliably by factual accuracy alone.

5.2 RQ2: Can the LLM judge identify factual labels and agent roles?

RQ2 asks whether the LLM judge can classify factual claims and agent roles reliably. The results are mixed. The judge performed moderately well at predicting FEVER claim labels, but performed poorly at detecting deceptive agents.

For factual claim prediction, judge claim accuracy was around 68–72% across the main comparisons. In the main debate comparison, judge claim accuracy increased slightly from 68.4% in prompting-only runs to 70.6% in FEVER-based RAG runs. In the matched evidence comparison, judge claim accuracy decreased slightly from 71.9% to 70.6%. These results suggest that evidence access did not produce a large or consistent improvement in the judge’s factual classification performance. Table 3 summarizes the main outcome metrics across the main debate comparison and the matched evidence comparison.

Table 3: Main outcome metrics across comparison groups and evidence conditions.

Comparison	Evidence condition	Runs	Judge claim accuracy	Final answer accuracy	Agent-role accuracy
Main debate	Prompting-only	320	68.4%	42.8%	53.1%
Main debate	FEVER-based RAG	320	70.6%	51.9%	53.1%
Matched comparison	Prompting-only	160	71.9%	48.8%	50.6%
Matched comparison	FEVER-based RAG	160	70.6%	53.1%	50.6%

The more important result concerns judge agent-role accuracy. In the main debate comparison, overall agent-role accuracy was 53.1% in both prompting-only and FEVER-based RAG runs. Since the role classification task is binary, this is close to chance. In the matched evidence comparison, agent-role accuracy was similarly low: 50.6% in both evidence conditions.

Overall accuracy also hides a strong prediction bias. The judge mostly predicted that agents were truth-oriented. In the main prompting-only comparison, the judge correctly identified truth-oriented agents 93.8% of the time, but correctly identified deceptive agents only 12.5% of the time. Under FEVER-based RAG, truth-oriented detection increased to 95.6%, while deceptive detection decreased to 10.6%. The matched evidence comparison showed the same pattern: deceptive detection was only 8.8% in both evidence conditions.

Figure 2 visualizes the same role-specific imbalance: truth-oriented detection remains high, while deceptive detection remains low across all evidence conditions.

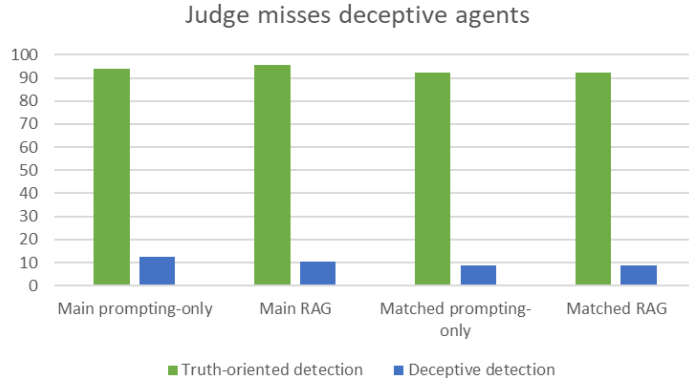


Fig. 2: Role-specific judge detection accuracy. The judge identifies truth-oriented agents at a high rate but misses most deceptive agents, showing a strong prediction bias toward the truth-oriented label.

This directly answers the behavioral part of RQ2. The judge was not reliable at detecting deceptive role-conditioned behavior. Its main failure was not random error, but a strong bias toward classifying transcripts as truth-oriented. This means that the judge appeared reliable on truth-oriented agents while missing most deceptive agents.

This result is important because deceptive behavior in the experiment is often expressed through emphasis, omission, selective interpretation, or persuasive framing rather than direct contradiction. The judge therefore needed to recognize not only whether a transcript sounded factual, but whether it used evidence and argumentation in a misleading way. The results suggest that the judge was weak at this behavioral classification task.

5.3 RQ3: Does FEVER-based RAG improve factual grounding or judge evaluation?

RQ3 asks how stable the judge classifications are across evidence conditions and scenario groups. More broadly, it asks whether FEVER-based RAG improves factual grounding and whether this improvement transfers to judge evaluation.

The clearest effect of FEVER-based RAG appears in final answer accuracy. In the main debate comparison, RAG improved final answer accuracy from 42.8% to 51.9%. In the matched evidence comparison, RAG improved final answer accuracy from 48.8% to 53.1%. This means that giving evidence to the debate agent helped the agent reach the correct FEVER label more often.

However, this improvement did not transfer to agent-role detection. Judge agent-role accuracy stayed almost unchanged across evidence conditions. In the main debate comparison, agent-role accuracy was 53.1% with and without RAG. In the matched evidence comparison, it was 50.6% with and without RAG. This shows that improving the agent’s evidence access helped factual answering more than behavioral role classification.

The same pattern appears in the behavioral scores. FEVER-based RAG improved judge-rated evidence behavior. In the main debate comparison, evidence faithfulness increased from 4.23 to 4.75, unsupported claims decreased from 1.38 to 0.94, and cherry-picking decreased from 0.42 to 0.32. The matched evidence comparison showed the same trend: evidence faithfulness increased from 4.18 to 4.79, unsupported claims decreased from 1.30 to 0.88, and cherry-picking decreased from 0.44 to 0.24. Tables 4 and 5 report the judge-rated behavioral scores across evidence conditions.

Table 4: Judge-rated factual and evidence scores across evidence conditions.

Comparison	Condition	Factual accuracy	Evidence faithfulness	Unsupported claims
Main debate	Prompting-only	3.35	4.23	1.38
Main debate	FEVER-based RAG	3.62	4.75	0.94
Matched comparison	Prompting-only	3.54	4.18	1.30
Matched comparison	FEVER-based RAG	3.67	4.79	0.88

Table 5: Judge-rated rhetorical and contradiction scores across evidence conditions.

Comparison	Condition	Cherry-picking	Persuasiveness	Contradiction with evidence
Main debate	Prompting-only	0.42	4.29	45.6%
Main debate	FEVER-based RAG	0.32	3.97	50.6%
Matched comparison	Prompting-only	0.44	4.31	—
Matched comparison	FEVER-based RAG	0.24	3.89	—

These results support a narrower interpretation of RAG. FEVER-based RAG improved factual grounding: answers became more tied to supplied evidence, contained fewer unsupported claims, and reached the correct FEVER label more often. However, RAG did not solve behavioral evaluation. The judge still failed to detect most deceptive agents.

Contradiction with evidence did not clearly improve in the main debate comparison. It increased from 45.6% to 50.6% under RAG. This result should be interpreted cautiously, because explicit evidence may make contradictions easier for the judge to notice. Still, it shows that evidence access does not automatically eliminate inconsistent or misleading argumentation. A response may cite evidence while still drawing an incorrect, selective, or overconfident conclusion from it.

Average persuasiveness also decreased under RAG. In the main debate comparison, persuasiveness decreased from 4.29 to 3.97. In the matched evidence comparison, it decreased from 4.31 to 3.89. One possible explanation is that evidence-constrained outputs are less rhetorically forceful than prompting-only outputs. Another possibility is that the judge rewards confident free-form argumentation as persuasive. Either way, this result reinforces the need to separate factual quality from rhetorical appeal.

The answer to RQ3 is therefore: FEVER-based RAG improved final answer accuracy and evidence-grounding indicators, but it did not improve the judge’s ability to identify deceptive role-conditioned behavior.

5.4 Label-specific factual performance

The label-specific analysis explains where RAG helped and where it failed. This analysis is relevant to RQ2 because it evaluates judge claim prediction, and to RQ3 because it shows how evidence access affects different FEVER labels.

RAG improved performance on decisive labels. In the main debate comparison, judge claim accuracy on supported claims increased from 84.0% to 100.0%, and accuracy on refuted claims increased from 87.5% to 94.2%. The matched evidence comparison showed the same pattern: supported claims reached 100.0%, and refuted claims reached 95.3% under RAG.

However, performance on not enough information claims became worse. In the main debate comparison, judge accuracy on not enough information dropped from 30.0% to 13.0%. In the matched evidence comparison, it dropped from 39.6% to 8.3%. Table 6 shows that the improvement under FEVER-based RAG is concentrated on supported and refuted claims, while not enough information performance decreases.

Table 6: Judge claim accuracy by FEVER label.

Condition	Supported	Refuted	Not enough information
Main debate: prompting-only	84.0%	87.5%	30.0%
Main debate: FEVER-based RAG	100.0%	94.2%	13.0%
Matched comparison: prompting-only	85.4%	85.9%	39.6%
Matched comparison: FEVER-based RAG	100.0%	95.3%	8.3%

This result shows that RAG made the system better at recognizing claims that could be resolved as supported or refuted, but worse at recognizing evidential insufficiency. One explanation is that the presence of an evidence block may encourage the model or judge to assume that the claim should have a decisive answer. In other words, evidence access may make the system more decisive without making it better at uncertainty recognition.

5.5 RQ4: How do accuracy and confidence change over debate turns?

RQ4 asks how factual accuracy, contradiction, confidence, and rhetorical behavior change over debate turns. The turn-level results show that debate did not produce continuous improvement.

In the main prompting-only comparison, final-answer accuracy was 60.3% on Turn 1, 35.3% on Turn 2, 56.3% on Turn 3, and 42.8% on Turn 4. Under FEVER-based RAG, the corresponding values were 67.8%, 39.1%, 59.4%, and 51.9%.

RAG improved accuracy at every turn, but the trajectory was not monotonic. Figure 3 visualizes this turn-level pattern.

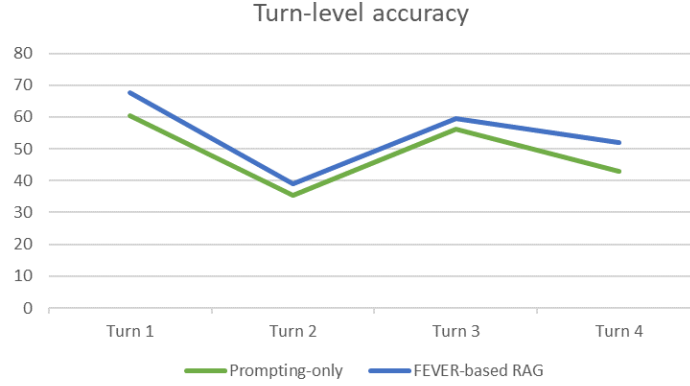


Fig. 3: Turn-level final answer accuracy across the four debate turns. FEVER-based RAG improves accuracy at every turn, but accuracy does not increase monotonically over the debate.

These results suggest that debate can introduce instability. Later turns do not automatically improve factual classification. In both evidence conditions, an earlier turn sometimes had higher accuracy than the final turn. This may happen because later turns respond to the previous generated text rather than only to the original claim and evidence. Debate history can therefore be useful context, but it can also become a source of drift.

Judge confidence was also weakly calibrated. In the main prompting-only comparison, mean judge confidence was 0.795 on correct predictions and 0.782 on incorrect predictions. The difference is small, meaning that the judge was almost as confident when wrong as when correct. Confidence therefore did not provide a reliable signal of correctness in this setup.

The answer to RQ4 is therefore: debate did not produce steady improvement over turns, and judge confidence did not clearly separate correct from incorrect predictions. This supports a cautious interpretation of LLM debate. Multi-turn debate can expose behavior and produce analyzable dynamics, but it should not be assumed to guarantee better reasoning or reliable self-assessment.

6 Discussion and Limitations

6.1 Discussion of the research questions

The four research questions point to the same overall interpretation. RQ1 showed that truth-oriented and deceptive role conditions produced different behavioral profiles, especially in persuasiveness and cherry-picking, but not in a simple “truthful equals correct, deceptive equals incorrect” way. RQ2 showed that the judge was moderately successful at factual claim prediction but unreliable at detecting deceptive agents, mainly because it strongly favored the truth-oriented

label. RQ3 showed that FEVER-based RAG improved final answer accuracy and evidence-grounding indicators, but did not improve agent-role detection. RQ4 showed that debate accuracy did not improve steadily over turns and that judge confidence was weakly calibrated. Together, these results suggest that factual grounding and deception detection should be treated as related but distinct evaluation problems.

6.2 Implications

These findings suggest that LLM-based evaluation should not rely on a single aggregate score. A judge may perform reasonably on factual labels while still failing to detect misleading framing or selective evidence use. For this reason, evaluation reports should include role-specific detection accuracy, label-specific performance, confusion matrices, and confidence calibration. The results also support a cautious view of debate-based AI safety: debate can create richer transcripts for evaluation, but it can also produce fluent and persuasive misleading arguments. For RAG systems, retrieval should therefore be paired with explicit checks for evidence sufficiency and selective use, especially when the correct answer may be not enough information.

6.3 Limitations

This study has several limitations. First, it studies role-conditioned debate simulations rather than deployed autonomous agents. The agents are prompted instances of the same base model inside a fixed pipeline; they do not have persistent memory, independent long-term goals, or autonomous tool-use strategies. The results should therefore be interpreted as evidence about prompt-conditioned debate behavior, not autonomous agents in real-world deployment.

Second, the same local Llama-family setup is used for the main debater and judge. This limits generalization across models and may introduce shared biases between generator and evaluator, such as a preference for fluent and confident text. Future work should compare multiple generator models and independent judge models.

Third, the study does not include human evaluation. FEVER labels make factual accuracy automatically checkable, but behavioral scores such as cherry-picking, persuasiveness, and misleading framing are produced by the LLM judge. A stronger design would include a small human-labeled validation set for deceptive or misleading behavior.

Finally, the dataset and evidence setting are controlled but narrow. The experiments use a small FEVER sample and FEVER-constrained RAG rather than open-world retrieval. This supports internal comparison, but does not capture the full complexity of real misinformation, noisy retrieval, conflicting sources, or claims that do not fit cleanly into supported, refuted, or not enough information. The analysis is also mostly descriptive, so the strongest conclusions are based on repeated patterns rather than broad statistical generalization.

7 Conclusion

This thesis investigated whether truth-oriented and deceptive role-conditioned LLM agents produce different behavior in controlled multi-turn factual debates, and whether an LLM judge can identify both the factual status of the claim and the role condition behind the generated transcript. The experiments compared prompting-only debate with FEVER-based retrieval-augmented generation, where RAG was applied to the debate agent by giving it controlled FEVER evidence during generation.

The results show that FEVER-based RAG improves the factual performance of the debate agent. Final answer accuracy increased under RAG, and the judge-rated behavioral indicators also suggested stronger evidence grounding: higher evidence faithfulness, fewer unsupported claims, and less cherry-picking. These grounding scores should be interpreted as LLM-judge assessments rather than human-validated annotations. Even with that caveat, the pattern suggests that giving the agent controlled evidence made its answers more tied to the available evidence.

However, this improvement in factual grounding did not transfer to deceptive-role detection. The judge’s factual claim prediction and its agent-role prediction behaved differently. The judge was moderately successful at predicting FEVER labels, especially for supported and refuted claims, but it was poor at identifying deceptive agents. Its predictions showed a strong tendency toward the truth-oriented label when compared with the known experimental role conditions. This is a prediction bias relative to the experiment design, not a claim about human judgment. The result means that the implemented LLM judge should not be treated as a reliable detector of deceptive role-conditioned behavior.

The label-specific results further show that uncertainty recognition remains a major weakness. FEVER-based RAG improved performance on supported and refuted claims, but performance on not enough information claims became worse. This was a weakness in factual label prediction: the system became better at decisive labels, but worse at recognizing when the available evidence was insufficient. This suggests that evidence access can make an LLM system more decisive without making it better calibrated about uncertainty.

Overall, the thesis supports the conclusion that factual grounding and deception detection are related but distinct evaluation problems. A system can become better at using evidence and still fail to detect selective framing, omission, or misleading emphasis. This matters for LLM-based evaluation because a factual correctness benchmark alone does not test whether an evaluator can recognize deceptive or strategically misleading argumentation.

The practical lesson is that LLM evaluation pipelines should be validated on the exact behavior they are expected to detect. If the goal is factual verification, gold claim labels are useful. If the goal is deception detection, factual labels are not enough. Evaluators also need to be tested on selective evidence use, misleading framing, uncertainty recognition, and role-specific failure cases.

References

1. Aly, R., Guo, Z., Schlichtkrull, M.S., Thorne, J., Vlachos, A., Christodoulopoulos, C., Cocarascu, O., Mittal, A.: The Fact Extraction and VERification Over Unstructured and Structured Information (FEVEROUS) shared task. In: Proceedings of the Fourth Workshop on Fact Extraction and VERification (FEVER), pp. 1–13. Association for Computational Linguistics (2021). <https://doi.org/10.18653/v1/2021.fever-1.1>
2. Bavaresco, A., Bernardi, R., Bertolazzi, L., Elliott, D., Fernández, R., Gatt, A., Ghaleb, E., Giulianelli, M., Hanna, M., Koller, A., Martins, A.F.T., Mondorf, P., Neplenbroek, V., Pezzelle, S., Plank, B., Schlangen, D., Suglia, A., Surikuchi, A.K., Takmaz, E., Testoni, A.: LLMs instead of human judges? A large scale empirical study across 20 NLP evaluation tasks. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Volume 2: Short Papers, pp. 238–255. Association for Computational Linguistics (2025). <https://doi.org/10.18653/v1/2025.acl-short.20>
3. Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D.M., Wu, J., Winter, C., et al.: Language models are few-shot learners. *Advances in Neural Information Processing Systems* 33, 1877–1901 (2020)
4. Deriu, J., Rodrigo, A., Otegi, A., Echegoyen, G., Rosset, S., Agirre, E., Cieliebak, M.: Survey on evaluation methods for dialogue systems. *Artificial Intelligence Review* 54, 755–810 (2021). <https://doi.org/10.1007/s10462-020-09866-x>
5. Du, Y., Li, S., Torralba, A., Tenenbaum, J.B., Mordatch, I.: Improving factuality and reasoning in language models through multiagent debate. *arXiv:2305.14325* (2023)
6. Hagendorff, T.: Deception abilities emerged in large language models. *Proceedings of the National Academy of Sciences* 121(24), e2317967121 (2024). <https://doi.org/10.1073/pnas.2317967121>
7. Jeong, S., Choi, Y., Kim, J., Jang, B.: Tool-MAD: A multi-agent debate framework for fact verification with diverse tool augmentation and adaptive retrieval. *arXiv:2601.04742* (2026)
8. Jiang, Y., Bordia, S., Zhong, Z., Dognin, C., Singh, M., Bansal, M.: HoVer: A dataset for many-hop fact extraction and claim verification. In: Findings of the Association for Computational Linguistics: EMNLP 2020, pp. 3441–3460. Association for Computational Linguistics (2020). <https://doi.org/10.18653/v1/2020.findings-emnlp.309>
9. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems* 33, 9459–9474 (2020)
10. Lin, S., Hilton, J., Evans, O.: TruthfulQA: Measuring how models mimic human falsehoods. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, Volume 1: Long Papers, pp. 3214–3252. Association for Computational Linguistics (2022). <https://doi.org/10.18653/v1/2022.acl-long.229>
11. Liu, J., Song, Y., Xiao, Y., Zheng, M., Tjautja, L., Borg, J.S., Diab, M., Sap, M.: Synthetic Socratic debates: Examining persona effects on moral decision and persuasion dynamics. *arXiv:2506.12657* (2025)
12. Ma, J., Hu, L., Li, R., Fu, W.: LoCal: Logical and causal fact-checking with LLM-based multi-agents. In: Proceedings of the ACM Web Conference 2025 (WWW)

- '25), pp. 1614–1625. Association for Computing Machinery (2025). <https://doi.org/10.1145/3696410.3714748>
13. Marioriyad, A., Nouri, A., Rohban, M.H., Baghshah, M.S.: Lying to win: Assessing LLM deception through human-AI games and parallel-world probing. ICLR 2026 Workshop on Agents in the Wild. arXiv:2603.07202 (2026)
 14. Ning, Y., Lin, X., Fang, F., Cao, Y.: MAD-Fact: A multi-agent debate framework for long-form factuality evaluation in LLMs. arXiv:2510.22967 (2025)
 15. Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C.L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Aspell, A., Welinder, P., Christiano, P., Leike, J., Lowe, R.: Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems* 35, 27730–27744 (2022)
 16. Thorne, J., Vlachos, A., Christodoulopoulos, C., Mittal, A.: FEVER: A large-scale dataset for fact extraction and verification. In: *Proceedings of NAACL-HLT 2018*, pp. 809–819. Association for Computational Linguistics (2018)
 17. Wang, H., Prasad, A., Stengel-Eskin, E., Bansal, M.: Retrieval-augmented generation with conflicting evidence. arXiv:2504.13079 (2025)
 18. Yang, S., Yu, Z., Ying, Z., Dai, Y., Wang, G., Lan, J., Xu, J., Li, J., Ngai, E.C.H.: RAMA: Retrieval-augmented multi-agent framework for misinformation detection in multimodal fact-checking. arXiv:2507.09174 (2025)
 19. Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., Stoica, I.: Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. arXiv:2306.05685 (2023)